

AI TOOLS EXPLORATION

Tool Used

ChatGPT

How I used it

I used ChatGPT as an AI learning and development assistant throughout this project. It helped me understand Python, Pandas, NumPy, Matplotlib, Scikit-learn, TF-IDF, model evaluation, and Streamlit.

Coding assistance

- Understanding Python syntax and concepts
- Understanding Pandas and NumPy operations
- Debugging Python errors
- Understanding TF-IDF
- Understanding classification metrics
- Understanding confusion matrices
- Building the machine-learning pipeline
- Creating the Streamlit application
- Debugging the application

Research/Learning assistance

ChatGPT was used to explain unfamiliar concepts in simple language and provide examples that helped me understand the underlying concepts rather than simply copying code.

Productivity assistance

AI assistance helped reduce the time required to troubleshoot errors, organize the project workflow, generate documentation ideas, and structure the project.

RESPONSIBLE USE OF AI

AI tools were used as learning and development assistants. I reviewed, tested, and modified the generated suggestions before incorporating them into the project. Errors and unexpected results were investigated and corrected during development.

The final model was evaluated using a test dataset rather than assuming that AI-generated code would automatically produce correct results.